

Heatless desiccant air dryers

WHD series

Where a +3 °C dew point is not dry enough, WHD heatless dryers pass compressed air through a bed of activated alumina or molecular sieve. One tower dries while the other regenerates with a small flow of purge air, switching every few minutes.



Capacity	Pressure dew point	Working pressure	Inlet temperature
300 to 2000 CFM	–20 °C (alumina) · –40 °C (molecular sieve)	7 to 12.5 bar g	45 °C

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[Specifications online](#)

Layout edition 2026-10 · Product-range information. Confirm the selected model and duty with Win Equipments before ordering.

Models & rating conditions

Rating conditions

Dew point rating	At 38 °C inlet and 7 bar g	Design pressure	13.2 bar g
Cycle time	10 min	Power supply	220 V AC ± 5%, 1 Ph, 50 Hz

Capacity & operating data

Model	Flow rate CFM	Flow rate m³/hr	Desiccant options	Electrical V / Ph / Hz
WHD 030	300	510	A / M	220 / 1 / 50
WHD 040	400	680	A / M	220 / 1 / 50
WHD 050	500	850	A / M	220 / 1 / 50
WHD 060	600	1020	A / M	220 / 1 / 50
WHD 075	750	1275	A / M	220 / 1 / 50
WHD 100	1000	1700	A / M	220 / 1 / 50
WHD 125	1250	2125	A / M	220 / 1 / 50
WHD 150	1500	2550	A / M	220 / 1 / 50
WHD 200	2000	3400	A / M	220 / 1 / 50

Construction, dimensions & connections

Model	L × W × H mm	In / out NB	Weight kg
WHD 030	1450 × 1100 × 2500	1½"	450
WHD 040	1450 × 1200 × 2500	2"	630
WHD 050	1650 × 1250 × 2300	3"	720
WHD 060	1650 × 1250 × 2500	3"	910
WHD 075	2100 × 1400 × 2300	4"	1100
WHD 100	2100 × 1400 × 2500	4"	1200
WHD 125	2100 × 1400 × 2800	6"	1500
WHD 150	2300 × 1500 × 2300	6"	2100
WHD 200	2300 × 1500 × 2700	6"	2400

- A: activated alumina, pressure dew point –20 °C at 38 °C and 7 bar g working pressure.
- M: molecular sieves, pressure dew point –40 °C at 38 °C and 7 bar g working pressure.
- Design pressure 13.2 bar g. Standard supply 220 V AC ± 5%, 1 Ph, 50 Hz.
- Model code: WHD + capacity code (e.g. WHD 100 = 1000 CFM) + desiccant type.

How it works & construction

How it works

A heatless desiccant dryer removes water vapour by adsorption onto a desiccant bed. No heater is used: a small part of the dried air regenerates the off-line tower.

1 Pre-filter

A pre-filter removes free water, oil and dust scale that would damage the desiccant.

3 Purge

A small flow of dry air is expanded through the off-line tower and exhausted through a muffler, stripping the moisture it holds.

5 After-filter

An after-filter removes desiccant dust before the air reaches your equipment.

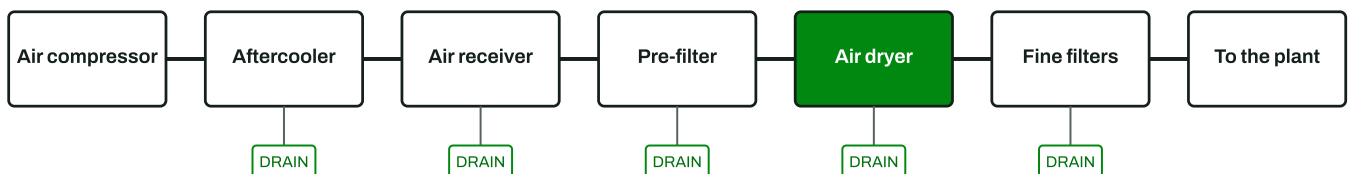
2 Adsorb

Wet air flows up through the on-line tower; the desiccant bed adsorbs water vapour. Low velocities stop bed fluidisation and dusting.

4 Switch

The timer and solenoid valves switch towers on a fixed cycle so one tower is always drying.

Typical compressed air installation



Typical arrangement only. Confirm filtration sequence and condensate handling for your application.

Construction & features

Controls

- Power-on light and on-off switch
- Accurate motorised cam timer
- PLC and cycle-failure indication (optional)

Valves

- Non-lubricated inlet control valves
- Purge flow valve to adjust purge rate
- Pressure relief valve on each tower

Quiet running

- Heavy-duty purge exhaust mufflers

Desiccant bed

- Large bed for sufficient contact time
- Sized to save heat of adsorption, minimising purge air
- Bulk desiccant: no special cartridges

Frame

- Structural steel frame with floor stand
- Lifting lugs
- Optional factory-mounted pre- and after-filters

Options

- Higher pressures and lower dew points on request

Selection & applications

Applications: Pharmaceutical · Electronics · Instrument air · Food packaging · Paint lines · Outdoor air lines in cold conditions

Alumina or molecular sieve: which do I need?

Activated alumina gives a -20°C pressure dew point; molecular sieve gives -40°C (both at 38°C inlet and 7 bar g). Choose molecular sieve when your process needs the lower dew point.

What does heatless mean?

The off-line tower is regenerated with a small flow of dry purge air instead of a heater, so there is no heater to maintain.

Why does a desiccant dryer need filters?

A pre-filter keeps free water and dust off the desiccant, a zero-oil filter stops compressor oil from coating it, and an after-filter catches desiccant dust before it reaches your equipment.

What working pressure is supported?

Standard WHD dryers work from 7 to 12.5 bar g with a design pressure of 13.2 bar g. Higher pressures can be offered on request.

Confirm your selection

Send the model, required capacity, operating temperatures, pressure and site conditions. For a model-specific datasheet or GA drawing, include your installation requirement.

Win Equipments

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Catalogue values are for product selection. Confirm duty, materials, connections and final dimensions with Win Equipments before ordering. Layout edition 2026-10.